KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME:

TEACHER:

MATHEMATICS

Year 9 5.2

Yearly Exam

2026

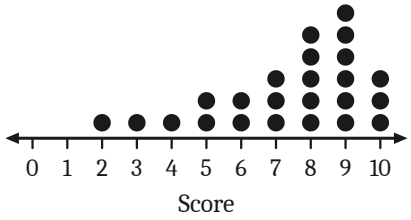
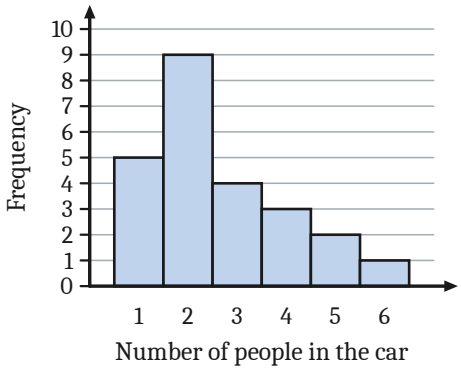
Instructions

1. Reading time: 5 minutes
2. Working time: 90 minutes
3. Attempt all questions
4. Show ALL working where needed
5. A calculator is permitted

Sections and Marks

Section 1 - Data	/17
Section 2 - Surface area and volume	/22
Section 3 - Mixed Topics	/33
Total marks	**/72**

Section 1: Data

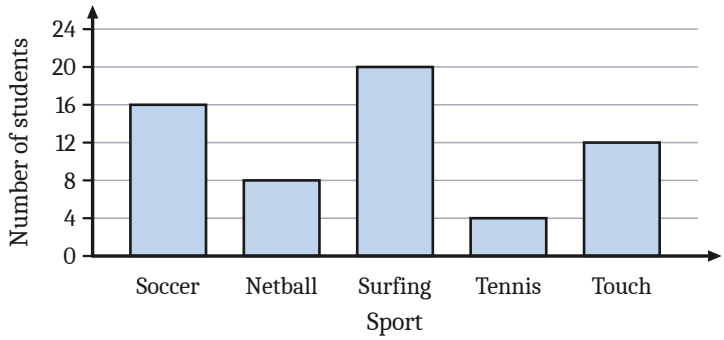
	Multiple Choice:	Mark
1	Which of the following is **numerical continuous** data? A. The number of goals in a hockey match B. The colours of cars in a car park C. Shirt sizes (S, M, L, XL) D. The time taken to swim 50 m	1
2	The dot plot shows the scores (out of 10) of 24 students in a spelling quiz. Which statement best describes the data? A. Positively skewed, clustered at 8 and 9 B. Negatively skewed, clustered at 8 and 9 C. Negatively skewed, clustered at 2 and 3 D. Symmetrical, clustered at 8 and 9 	1
3	The histogram shows the number of people in each of 24 cars that arrived at a beach car park. What is the mode? A. 2 B. 3.5 C. 6 D. 9 	1
4	A water polo team scored these numbers of goals in 10 games: 5, 7, 6, 8, 6, 21, 7, 5, 6, 8 Which measure of centre changes the most if the outlier, 21, is removed? A. The mode B. The median C. The mean D. All three change by the same amount	1

Section 1: Data (continued)

Short Answer:

	Question	Working Out	Mark
5	The stem-and-leaf plot shows the number of minutes that 11 students spent on homework last night. Stem | Leaf 1 | 2 5 8 2 | 1 4 4 7 3 | 0 3 5 6 Key: 2 | 4 = 24 minutes a. Find the mean. b. Find the median. Note: Mean = sum of the scores ÷ number of scores	a. b.	2 1

Worded Problems:

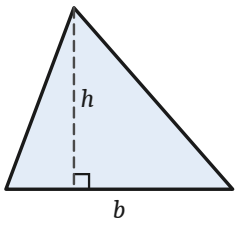
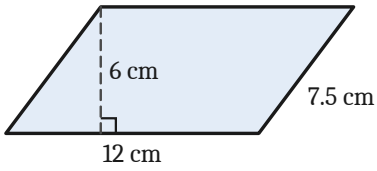
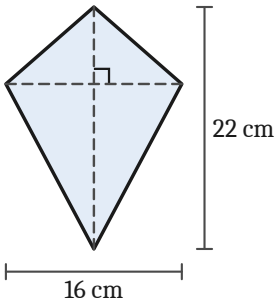
	Question	Working Out	Mark
6	Year 9 students each chose one sport for Friday afternoon. The column graph shows how many students chose each sport.  		
a	How many more students chose surfing than tennis?		1
b	How many students were surveyed altogether?		1
c	What percentage of the students chose surfing or tennis?		2

Section 1: Data (continued)

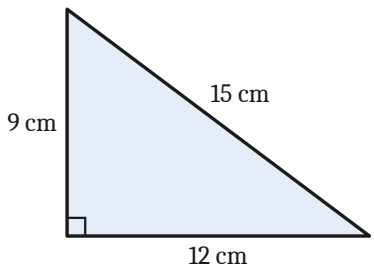
	Question	Working Out	Mark
7	The Kestrels and the Tigers are two netball teams. The back-to-back stem-and-leaf plot shows the number of goals each team scored in its 12 games this season. Note: Range = highest score − lowest score	Kestrels | Stem | Tigers 2 | 2 7 8 8 7 5 4 1 | 3 | 3 6 9 6 4 3 2 1 0 | 4 | 1 4 5 7 5 | 0 3 7 Key: 4 | 1 = 41 goals	
a	Find the median number of goals for each team.		2
b	Which team usually scores more goals? Give a reason.		1
c	Which team is more consistent? Find the range of each team to explain.		3
	Total		**/17**

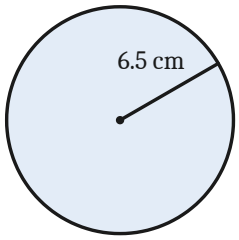
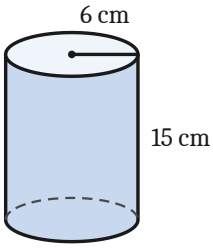
Section 2: Surface area and volume

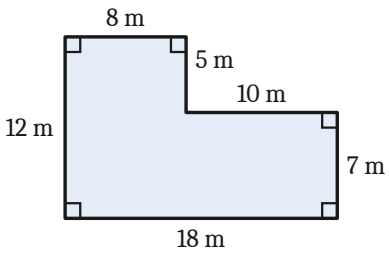
Multiple Choice:

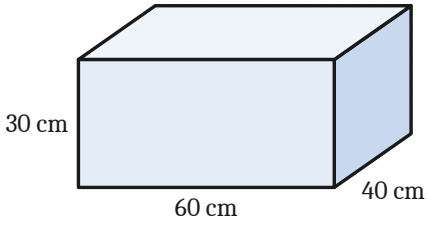
		Mark
1	A rainwater tank holds 4.6 kL of water. How many litres is this? A. 460 L B. 46 000 L C. 4600 L D. 0.0046 L	1
2	Select the correct formula for the area of a triangle. Note: b is the base and h is the perpendicular height A. $A = 2bh$ B. $A = bh$ C. $A = \frac{1}{2}bh$ D. $A = \frac{1}{2}(b + h)$ 	1
3	What is the area of this parallelogram? Note: Area of a parallelogram: $A = bh$ A. 90 cm^2 B. 36 cm^2 C. 39 cm^2 D. 72 cm^2 	1
4	What is the area of this kite? Note: Area of a kite: $A = \frac{1}{2}xy$ (x and y are the diagonals) A. 352 cm^2 B. 176 cm^2 C. 88 cm^2 D. 38 cm^2 	1

Short Answer:

	Question	Working Out	Mark
5	For this right-angled triangle, find:  a. its perimeter b. its area. Note: Area of a triangle: $A = \frac{1}{2}bh$	a. b.	1 1

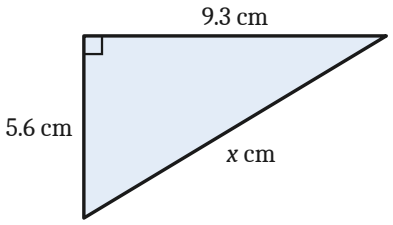
	Section 2: Surface area and volume (continued)		
	Question	**Working Out**	**Mark**
6	For this circle, find, correct to 1 decimal place:  a. its circumference b. its area. Note: Circumference: $C = 2\pi r$ Note: Area: $A = \pi r^2$	a. b.	1 1
7	This closed cylinder has a radius of 6 cm and a height of 15 cm.  a. Find its volume, correct to 1 decimal place. b. Find its surface area, correct to 1 decimal place. Note: Volume: $V = \pi r^2 h$ Note: Surface area: $SA = 2\pi r^2 + 2\pi rh$	a. b.	2 2

	Section 2: Surface area and volume (continued)		
	Worded Problems:		
	Question	**Working Out**	**Mark**
8	The plan shows an L-shaped lawn at a primary school. All of the corners are right angles. Note: Area of a rectangle: $A = lb$ 		
a	Garden edging is laid around the outside of the lawn. How many metres of edging are needed?		1
b	Find the area of the lawn.		3
c	New turf costs $12.40 per square metre. Find the cost of the turf for the whole lawn.		1

	Section 2: Surface area and volume (continued)		
	Question	**Working Out**	**Mark**
9	A closed wooden storage box is a rectangular prism 60 cm long, 40 cm wide and 30 cm high.  		
a	Find the volume of the box in cm^3. Note: Volume of a rectangular prism: $V = lwh$		1
b	How many litres does the box hold? Note: 1 L = 1000 cm^3		1
c	The outside of the box is to be painted. Find the surface area of the box. Note: Surface area = the total area of the 6 faces; area of a rectangle: $A = lb$		3
	Total		**/22**

Section 3: Mixed Topics (Chapters 1–8)

Multiple Choice:

		Mark
1	If $a = 5$ and $b = 3$, evaluate $2a^2 - b$. A. 7 B. 17 C. 47 D. 97	1
2	Find the value of x. Select the closest answer. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ A. 14.9 B. 10.9 C. 7.4 D. 3.7 	1
3	A skateboard is marked at \$145. In a sale it is discounted by 15%. What is the selling price? A. \$166.75 B. \$130.00 C. \$123.25 D. \$21.75	1
4	Simplify $4m^6 \times 3m^2$. A. $12m^8$ B. $7m^8$ C. $12m^{12}$ D. $12m^4$	1
5	Ella is a nurse. She earns an annual salary of \$81 640. How much is she paid per fortnight? Note: 1 year = 52 weeks = 26 fortnights = 12 months A. \$1570.00 B. \$3140.00 C. \$3401.67 D. \$6803.33	1

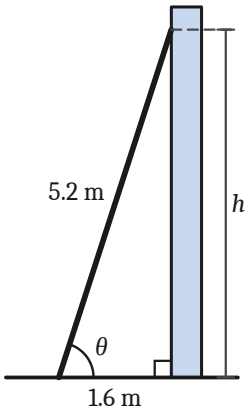
Short Answer:

	Question	Working Out	Mark
6	Simplify: a. $5a + 3b - 2a + b$ b. $4x \times 3y$ c. $18mn \div 6m$	a. b. c.	1 1 1

Section 3: Mixed Topics (Chapters 1–8) (continued)			
	Question	**Working Out**	**Mark**
7	Expand and simplify: $3(4x - 5) - 2(x - 6)$		2
8	Factorise: a. $6x + 15$ b. $y^2 - 8y$	a. b.	1 1
9	Liam invests \$7250 for 3 years at 4.2% p.a. simple interest. Calculate the interest he earns. Note: Simple interest: $I = Prn$		2
10	A surf shop buys a surfboard for \$480 and sells it for \$612. a. Find the profit. b. Find the profit as a percentage of the cost price. Note: Percentage profit = profit ÷ cost price × 100%	a. b.	1 1

	Section 3: Mixed Topics (Chapters 1–8) (continued)		
	Question	**Working Out**	**Mark**
11	Solve each equation. a. $5x - 8 = 27$ b. $7m - 2(m + 3) = 3m + 10$	a. b.	2 3
12	Jake sells cars. He is paid $520 per week plus 2.5% commission on his sales. Last week he sold $46 800 worth of cars. Calculate Jake's total pay for the week.		2
	Worded Problems:		
	Question	**Working Out**	**Mark**
13	Mia works in a bakery. Her normal rate of pay is $26.80 per hour for the first 38 hours each week. She is paid time-and-a-half for each hour after that. Last week Mia worked 44 hours.		
a	Calculate Mia's pay for the first 38 hours.		1
b	Calculate Mia's total pay for the 44 hours.		2

Section 3: Mixed Topics (Chapters 1–8) (continued)

	Question	Working Out	Mark
14	A gym charges a joining fee of \$60 plus \$15 per week. The total cost, C dollars, for w weeks is given by the formula $\boldsymbol{C = 60 + 15w}$		
a	Find the total cost for 12 weeks.		1
b	Jo paid \$300 altogether. Write an equation and solve it to find how many weeks Jo went to the gym.		2
15	A ladder 5.2 m long leans against a vertical wall. The foot of the ladder is 1.6 m from the wall, on level ground. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ Note: $\sin\theta = \frac{\text{opp}}{\text{hyp}}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}}$ 		
a	Find h, the height the ladder reaches up the wall, correct to 2 decimal places.		2
b	Find the angle θ that the ladder makes with the ground, correct to the nearest degree.		2
	Total		**/33**